

Title Header

- **Competitor / Developer:** Respect Kagiso MORUPANE
- **Institution:** Mopani TVET College
- **Build / Solution Name:** Project Vector Racer

Context & Problem Statement

Project Vector Racer is engineered to compete in the RoboRumble 2026 Autonomous Car Racing (Robo Grand Prix) category. The competition mandates strict physical and financial constraints that heavily influenced the vehicle's architecture. The vehicle must remain strictly under the 5 kg weight limit and fit within a 50 cm x 50 cm dimensional arena footprint, which was addressed by selecting a compact, lightweight 1:10 scale touring frame (total mass ~1.15 kg). The strict R 3,000.00 budget limit required careful component selection, utilizing an ESP32 microcontroller instead of a high-cost mini-PC, keeping the final build cost at R 2,990.00. Furthermore, the arena rules mandate immediate physical safety disarms; this is addressed through a dual-interlock system featuring a master physical battery disconnect and a hardware-level E-Stop button linked directly to a motor-disarm interrupt.

Solution Overview

Project Vector Racer is a 1:10 scale rear-wheel-drive autonomous race car designed to navigate a high-speed circuit without human intervention. It utilizes a dual-core ESP32-WROOM-32U microcontroller to process environmental data in real-time. Forward track layout is mapped using a TF-Luna Micro LiDAR sensor (8-meter range), while lateral wall tracking is handled by dual Sharp IR analog sensors. Steering is controlled via a 20kg.cm metal-gear digital servo using Ackerman geometry, and forward propulsion is driven by a 3650 3900KV sensor less brushless motor paired with a 60A ESC, governed by a custom Proportional-Integral-Derivative (PID) control algorithm.

BOM Summary

Below is the summarized list of core components. Please refer to the [Main Excel Bill of Materials \(Bill of Materials.xlsx\)](#) for the full itemized breakdown and supplier links.

Core Component	Purpose	Unit Cost
1:10 RWD Touring Chassis	Structural frame and mechanical drivetrain base	R 850.00
ESP32-WROOM-32U MCU	Primary processing, logic, and motor control	R 145.00

Core Component	Purpose	Unit Cost
TF-Luna Micro LiDAR	Long-range forward obstacle and turn detection	R 450.00
Sharp IR Sensors (x2)	Lateral wall tracking and boundary clearance	R 360.00
3650 Motor + 60A ESC	Brushless high-speed propulsion system	R 450.00
DS3218MG 20kg Servo	High-torque Ackerman steering actuation	R 290.00
2S 2200mAh LiPo Battery	Primary power source (7.4V nominal)	R 220.00

Mechanical Design Section

The vehicle's physical fabrication centres around modifying a commercial 1:10 scale RWD on-road touring chassis to support autonomous hardware. The base frame provides an adjustable 257 mm wheelbase and 190 mm track width, perfectly fitting the <50x50 cm arena rule.

To ensure the sensors function without interference, custom mounts were fabricated using 3D-printed PETG filament. PETG was chosen over PLA for its high impact resistance and thermal stability near the brushless motor. The TF-Luna LiDAR is elevated 100 mm above the main deck via a custom printed tower to eliminate floor-bounce signal degradation. The dual Sharp IR sensors are housed in custom brackets that lock them at a precise 15-degree toe-out angle to maximize lateral wall detection during cornering. The heavy 2S LiPo battery and 3650 motor are mounted low and shifted backward, achieving a 45/55 rear-biased weight distribution that maximizes traction on the drive wheels.

|<----- Overall Length: 257 mm ----->|

[Left IR Sensor]	[TF-Luna LiDAR]	[Right IR Sensor]	^
(15° Toe-out)	(100mm Tower)	(15° Toe-out)	
+-----+			
+-----+			
[0]		[0]	
	<===== [Tie Rod Linkage] =====>		
[0] (Front Wheel)		(Front Wheel)[0]	
+-----+			
	[DS3218MG Servo]		
	(Centerline)		Overall
			Width:
	[ESP32 Development Board]		190 mm
	& 5V Buck Converter		
	[7.4V 2S LiPo Battery]		
	(Low Center Floor)		

	[ESP32 Development Board]		190 mm
	& 5V Buck Converter		
	[7.4V 2S LiPo Battery]		
	(Low Center Floor)		
+-----+			
[0]		[0]	
	[60A Brushless ESC]		
	[Rear Axle]		
[0] (Rear Drive)		(Rear Drive)[0]	v
+-----+			
<----- Wheelbase: 257 mm ----->			

Side Elevation View & Height Dimensioning

|<----- 257 mm ----->|

|<----- Wheelbase: 257 mm ----->|

Side Elevation View & Height Dimensioning

|<----- 257 mm ----->|

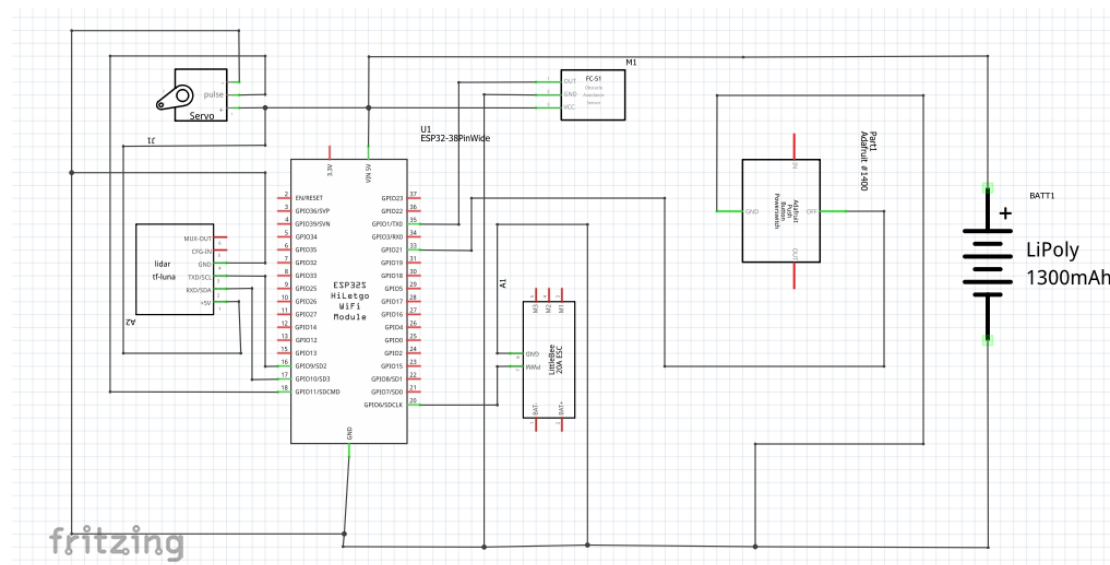
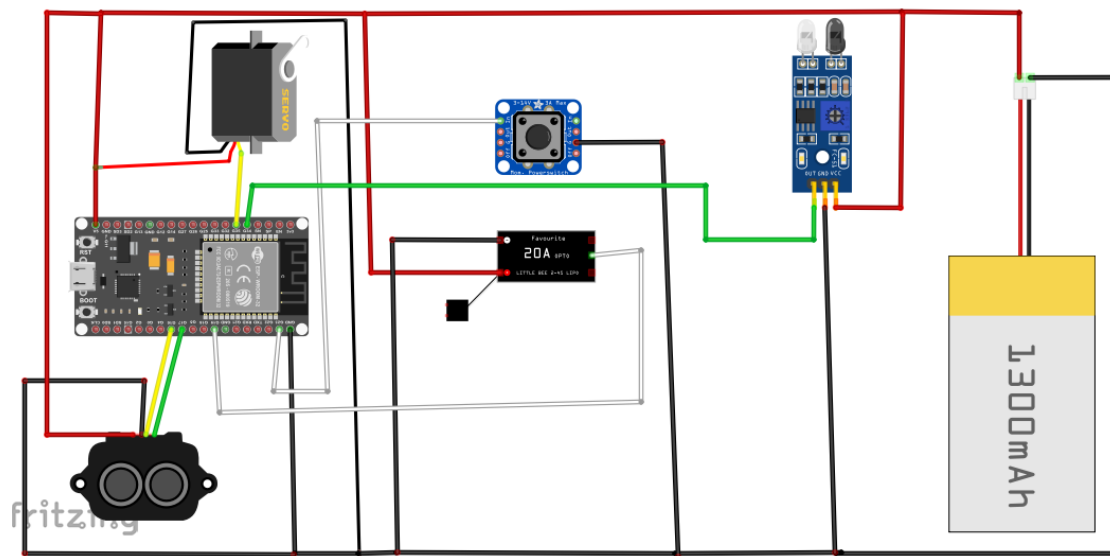
+-----+		--- 130 mm (Max Height)
TF-Luna LiDAR Tower		^
+-----+		
		100 mm Elevation
[Upper Deck]		v
+-----+		---
Left/Right IR Sensors ESP32 / Buck / ESC		^
+-----+		35 mm Elevation
[7.4V LiPo Battery]	(Chassis Floor)	
+-----+		---
(0)	(0)	^ 12 mm Ground Clearance
[Front Wheel]	[Rear Wheel]	v
=====		

Electronic Design Section

The hardware electrical system is divided into a dual-rail power hierarchy to isolate sensitive logic components from high-current motor noise.

- **Power Rail 1 (Unregulated 7.4V):** The 2S LiPo battery connects directly to the 60A ESC to supply high-current surges to the brushless motor. A heavy-duty 20A inline toggle switch acts as the master isolation switch.
- **Power Rail 2 (Regulated 5.0V):** An LM2596 DC-DC Buck Converter steps down the 7.4V battery voltage to a perfectly stable 5.0V (3A max), which safely powers the ESP32, LiDAR, IR sensors, and the steering servo.

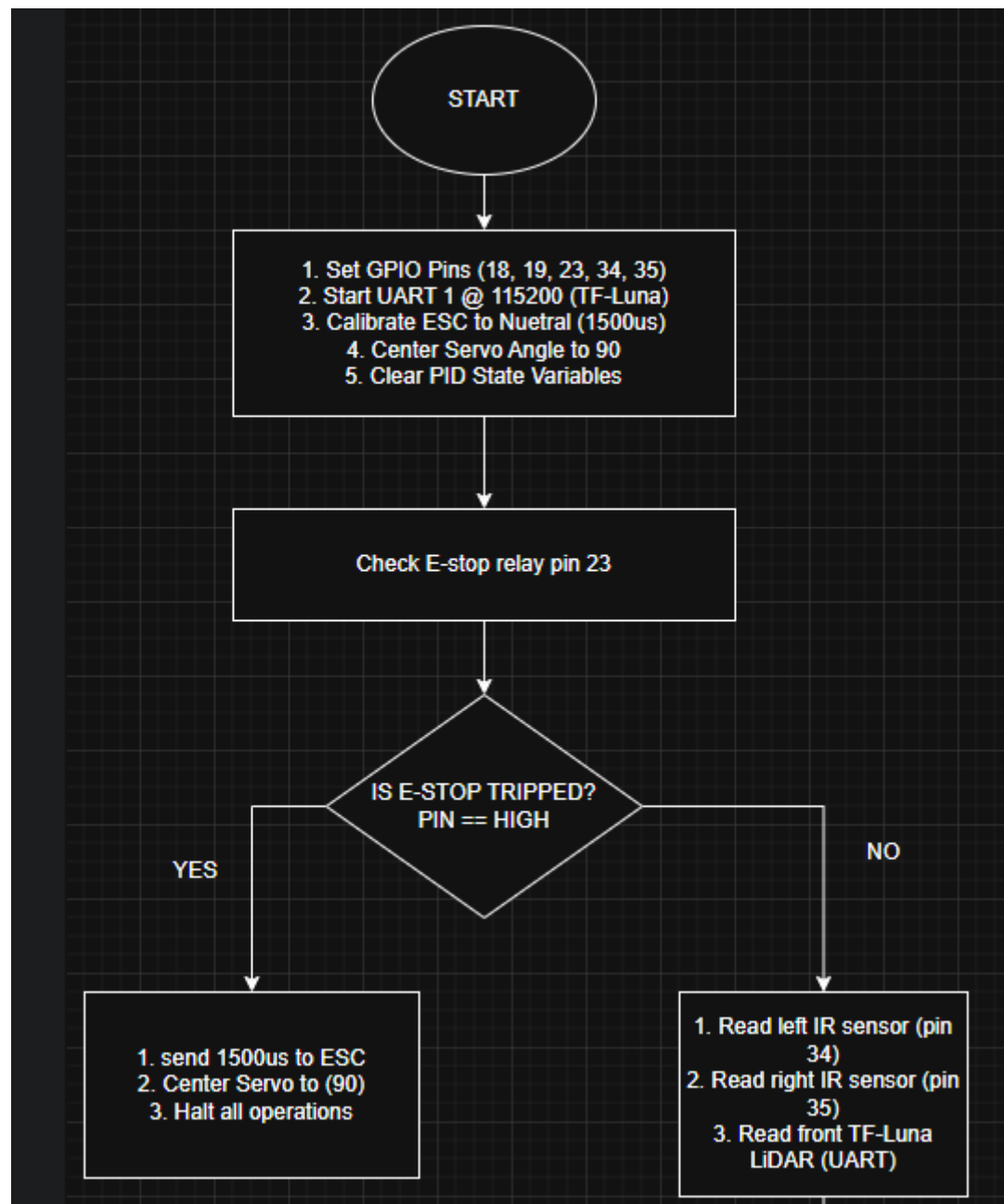
Mandatory E-Stop Integration: Per safety regulations, a red latching Emergency Stop button is wired between the ESP32's GPIO 23 and GND. When pressed, it pulls the pin LOW, triggering a hardware Level-1 Interrupt Service Routine (ISR) that instantly bypasses the main logic loop and forces the ESC PWM signal to a neutral 1500 microseconds, stopping the car dead regardless of sensor input.

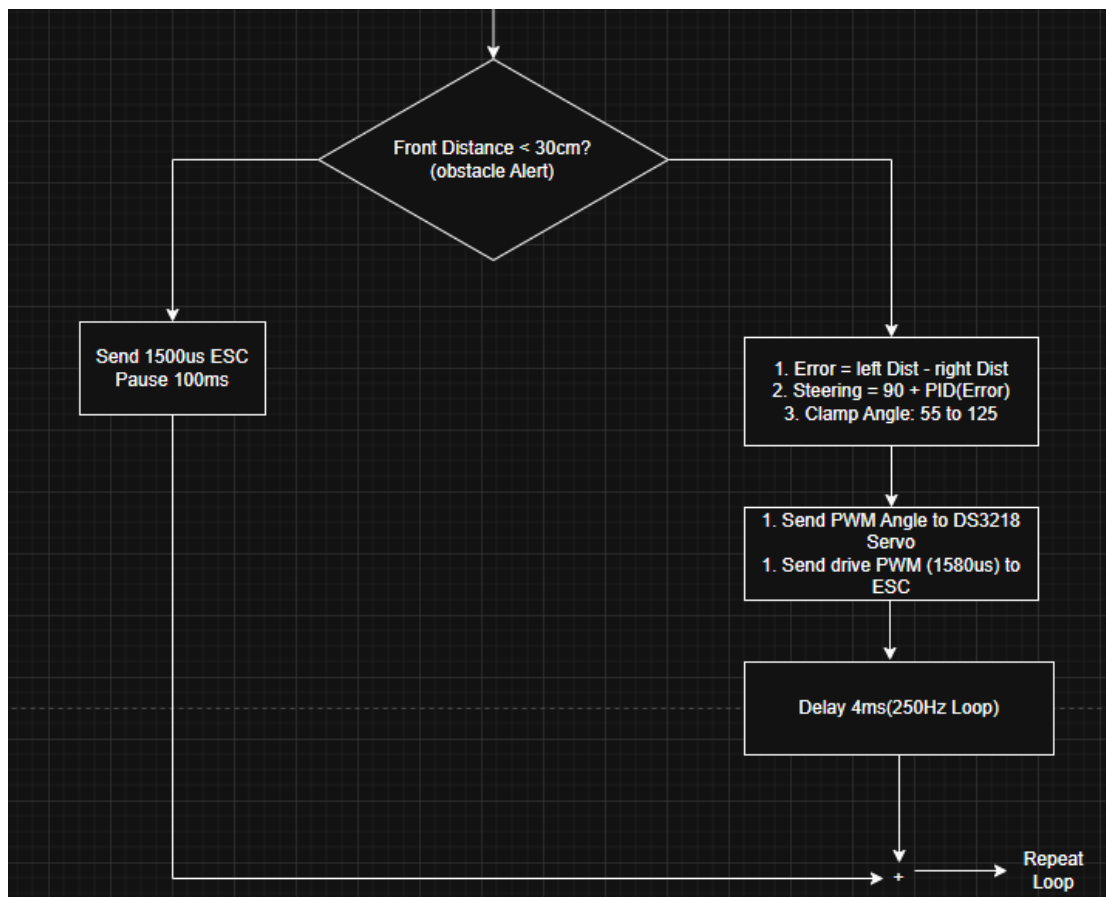


Programming & Framework Design Section

The software architecture leverages the ESP32's dual-core processor using a FreeRTOS framework to ensure zero-latency sensor processing and motor execution.

- **Task Isolation (FreeRTOS):**
 - **Core 0 (Perception Loop at 250 Hz):** Dedicated strictly to listening to UART packets from the LiDAR and reading the analog-to-digital converter (ADC) pins (GPIO 34 and GPIO 35) for the IR sensors.
 - **Core 1 (Control Loop at 100 Hz):** Takes the filtered distance variables from Core 0 and runs them through a Proportional-Integral-Derivative (PID) algorithm. The PID calculates the required steering angle error and adjusts the servo PWM signal, simultaneously dynamically throttling the ESC speed based on the forward LiDAR clearance.





Core Code Snippet (E-Stop ISR Execution):

// E-Stop Hardware Interrupt Initialization

```
const int ESTOP_PIN = 23;
```

```
volatile bool system_disarmed = false;
```

// Interrupt Service Routine (ISR) runs entirely independent of main loop

```
void IRAM_ATTR handleEmergencyStop() {
```

```
    // 1500us is the standard neutral/brake signal for RC ESCs
```

```
    ledcWrite(ESC_CHANNEL, 1500);
```

```
    ledcWrite(SERVO_CHANNEL, 90); // Center steering
```

```
    system_disarmed = true;    // Flag to halt all Core 0 & Core 1 PID calculations
```

```
}
```

```
void setup() {
```

```
    pinMode(ESTOP_PIN, INPUT_PULLUP);
```

```
    attachInterrupt(digitalPinToInterrupt(ESTOP_PIN),    handleEmergencyStop,
    FALLING);
```

```
}
```